A shorter prefix helps only when the pool is full

Qwen3-14B · one H200 · vLLM 0.11.1 · bf16

Sagnik Biswas

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Three checks

Same model, same GPU. Pool: **29.5 GiB** (12,102 blocks). One 32k decode is **5.0 GiB**. One block is 2.5 MiB.

- Clear the cache and the same prompt is computed again. Send it once more and the saved result is used.
- A shorter prompt uses less GPU memory while those 32k tokens are generated.
- Five long prompts do not fit. vLLM throws a running request out and starts it over. Five short prompts fit, so that does not happen.

Caching behaves

probe	cached / prompt	
cold	0 / 525	miss
identical replay	512 / 525	hit
one block changed	256 / 525	suffix misses
different question	64 / 617	shared prefix only
after cache clear	0 / 525	cleared
replay after clear	512 / 525	hit again

8,064-token prompt: 8,048 cached on replay, 0 after the clear.

A 77-token cut does nothing

525 vs 448 tokens, about 12 MiB. Caching off. Every request wrote 5 GiB.

requests	full	short	preemptions
1	324 s	324 s	0, 0
2	344 s	344 s	0, 0
4	389 s	388 s	0, 0
6	955 s	920 s	518, 479

Both arms finish at 6. The short arm ran second, so the 35 s is partly a warm engine.

6.2 GiB vs 5.2 GiB per job, caching off

Long prompt: 6.2 GiB per request. Five are 31.2 GiB, over the 29.5 GiB pool. Short prompt: 5.2 GiB. Five are 25.8 GiB, inside it.

run	used	free	preemptions	time
4 × long	24.9 GiB	4.6 GiB	0	7.1 min
4 × short	20.6 GiB	8.9 GiB	0	6.5 min
5 × long	29.5 GiB	42 MiB	807	30.7 min
5 × short	25.8 GiB	3.8 GiB	0	6.9 min

The 4-request gap is 4.3 GiB. The shorter prefix predicts 4.3 GiB.